

ALSu DCCT Test Setup & Procedure Checklist

This checklist guides the technician through the physical setup, instrument configuration, and software execution required to test the ALSu DCCT modules.

1. Pre-Test Preparation

- ☐ The testing laptop is connected to the local instrument network (ensure IPs `10.0.142.x` and `10.0.143.x` are reachable).
- ☐ The Python virtual environment is activated.
- ☐ Required packages from `requirements.txt` are installed.

2. Hardware Power-Up

Ensure all laboratory instruments are turned ON:

- ☐ Kepco Power Supply
- ☐ Rigol DP800 Series Power Supply
- ☐ Rigol DG4000 Series Signal Generator
- ☐ Tektronix DPO4000/MSO4000 Series Oscilloscope
- ☐ Keysight 34461A or Keithley 2100 Digital Multimeter (DMM)
- ☐ Differential Probes for Oscilloscope

3. Physical Connections & Wiring

- ☐ **GRAY** CT conductors are routed and secured to the **GRAY** terminal blocks.
- ☐ **BLUE** conductors are routed and secured to the **BLUE** terminal blocks.
- ☐ Conductors loop through the DCCT only once, and wire is at least 2" away from rear of DCCT Chassis (Past the yellow spacer sticks)

4. Manual Instrument Configuration

While the script handles most parameters, the following must be set manually prior to execution:

- ☐ **Kepco PSU Front Panel Configuration:**
 - ☐ VOLTAGE CONTROL Switch: **OFF**
 - ☐ CURRENT CONTROL Switch: **OFF**
 - ☐ *Note: MODE Switch and Potentiometer positions do not matter.*

- ☐ **Oscilloscope Probes:**
 - ☐ Differential probes are powered **ON**.
 - ☐ Differential probes are physically set to **x20 Attenuation**.

5. Software Execution

- ☐ Open a terminal/command prompt and run: `python main.py` .
- ☐ When prompted, enter your **Name** and confirm.
- ☐ When prompted, enter your **Life #** and confirm.
- ☐ Scan or manually enter the **DCCT Serial Number** (the script will capture the first 13 characters) and confirm.
- ☐ Acknowledge the on-screen connection confirmation prompts by pressing `Return` .

6. Testing & Review

- ☐ Allow the script to initialize the instruments and run the **FLT12 Fault 1 & 2 Tests**.
- ☐ If FLT12 tests fail, select whether to retry, restart with a new unit, or proceed to the next step.
- ☐ Allow the script to run the **DCCT Current Tests** (Phase, Frequency, and Voltage validation).
- ☐ Once completed, verify that the automated **PDF Test Report** opens on your screen.
- ☐ Verify that raw `.csv` and `.png` plot data were successfully saved in the new `Test_Data/DCCT_<SN>-<Timestamp>/raw_data/` directory.
- ☐ Affix a Green or Red 'QA' sticker to indicate the unit's pass or fail status
- ☐ Print reports as needed, and upload to Git repository at end of shift for archival